Application of Convolutional Neural Networks and Vision Transformers for the Multi-Class Classification of Infectious Respiratory Diseases from Chest Radiological Images

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Abstarct - Mild, easily accessible, and inexpensive, chest radiography is a valuable modality for screening infectious respiratory disease, but manual interpretation is time consuming and prone to inter-observer variability. The study evaluates five deep-learning models for five-class classification of frontal chest X-ray images from a public dataset of 32,687 images—including a baseline CNN, a CNN with transfer learning for the VGG19 and ResNet50 architectures, a custom CNN with depthwise-separable convolutions and residual connections with spatial attention, and a Vision Transformer (ViT) trained from scratch. Heterogeneity of images and a 2.4:1 class imbalance are taken care of with a preprocessing pipeline.On a fixed 6,540-image test split, VGG19, ResNet50 and the custom CNN reached macro F1-scores of 0.945, 0.937 and 0.925 with ROC-AUC of 0.995, 0.991 and 0.993, whereas the from-scratch ViT reached 0.786 and 0.951. About 2.5 M parameters are used by the custom CNN, which is around 15 times less than VGG19. The accuracy of the custom-CNN (0.97 vs 0.92) is not explained. Some models are apparently reacting to non-anatomical artefacts, as indicated by the Grad-CAM and Guided Grad-CAM plus attention-map analyses. This is a study on the image classifications and is not a clinical diagnostic system.

Keywords—chest X-ray, convolutional neural network, Vision Transformer, transfer learning, class imbalance, Grad-CAM, respiratory disease

I. INTRODUCTION

Infectious respiratory diseases like COVID-19, tuberculosis (TB) and viral pneumonia continue to pose a significant public-health challenge. Chest X-ray (CXR) imaging is a fast, inexpensive, readily available tool in resource-limited environments but is limited by the sensitivity of the observer, the workload of the observer during an outbreak and the different anatomical structures that overlap in a 2D projection. For selected tasks on the CXR dataset, deep convolutional networks demonstrated comparable performance to radiologists, and for large natural-image datasets, self-attention models like the Vision Transformer (ViT) were able to match or outperform CNNs.

These models are a healthcare-analytics challenge, rather than a vision challenge: class imbalance, provenance of the datasets, performance beyond accuracy, and interpretability are issues to be addressed when building a model for use in radiology, since the end user is a clinician. Previous systematic review of COVID-19 CXR models revealed that models may not be useful clinically due to methodological flaws and dataset biases [3] and the models may include confounders instead of the disease [4].

The approach used in many CXR studies on the dataset considered here is binary COVID-19 vs. non-COVID-19 labels, which ignores the problem of differential diagnosis; class imbalance is often omitted from the study discussions; accuracy is often reported solely, and visual explanations are rarely provided. In this paper, an imbalanced-aware training algorithm is used, and an entire set of metrics and matched interpretation analysis are provided for a five-class problem.

Its goals were to (i) characterise a public 5 class CXR dataset, (ii) design and inspect a pre-processing pipeline, (iii) implement global class-based weighting rather than resampling, (iv) fine-tune VGG19 and ResNet50 with two stages, (v) design a spatial-attention CNN, (vi) train a ViT from scratch, and (vii) compare all 5 models on one fixed test split, using accuracy, precision, recall, F1, ROC-AUC and top-2 accuracy as metrics, along with Grad-CAM and attention-map analysis.

II. LITERATURE REVIEW

A. CNNs, image enhancement and attention-enhanced CNNs.

On COVQU-20, a CXR dataset of 18,479 images across three classes, the five enhancement methods (histogram equalisation, CLAHE, image complement, gamma correction and BCET) were compared with several CNNs; gamma correction showed the best performance with an accuracy of 96.29% [5]. The same in the present dataset, as

the present dataset shares its curation lineage [6] which means that the enhancement choice may matter, while the present study opted for the most suitable CLAHE parameters for the dataset by visual inspection, and did not perform an accuracy ablation. Sitaula and Hossain designed an attention module for VGG-16 network for COVID-19 CXR classification [7] and Jain et al. pre-trained CNNs for COVID-19 CXR analysis [8] both mainly focus on COVID-19 detection and do not perform a five-way infectious-disease discrimination.

B. Inspiration from the past and the need for data

ImageNet transfer learning has been shown to allow ViTs to compete with CNNs for medical tasks, as reported by Matsoukas et al. [9], and self-supervised pretraining to improve the performance of ViTs for classification of CXR and dermatology images, as reported by Azizi et al. [10]. The authors of [11] investigated the impact of training data, augmentation and regularisation on the performance of ViT. The ViT-based methods for COVID-19 CXR consist of a multi-task model using a low-level feature corpus [12], the transfer-learned and explainable xViTCOS [13] and a ViT with radiomics [14]. Limited annotated data and data hunger were cited by medical-imaging survey as key problems for transformers [15]. Pretraining or transfer learning, which the present ViT did not use, is common to these studies.

C. Explainability and Class imbalance

The main families of long-tailed learning are class re-balancing, information augmentation and module improvement, where the imbalance here is a moderate one and loss weighting was used. To achieve the explainability, a survey on XAI in medical image analysis highlights the current prevalence of saliency methods and the requirement to be careful when validating [17]; and a benchmark on CXR showed that the saliency methods, such as the Grad-CAM [18], tended to localise pathologies less effectively than the human benchmark [19]. In conjunction with shortcut learning evidence [4], this encourages the viewpoint that heatmaps are hypotheses, not facts.

D. Research gap

The detection of COVID-19 or 3-class framing is common across the studies analyzed, while most of the studies that rely on transformer architectures employ pretraining, and only a few of the studies analyzed pair interpretability and transformer architectures on the same split. The current work presents an 5-class evaluation, a class-weighted training, a well-documented preprocessing procedure, a same-split comparison between CNN and the attention-CNN and ViT architectures, and an interpretability analysis. We cannot say that this mix is unprecedented, we looked at a limited range of reviews.

III. DATA SET & EXPLORATORY DATA ANALYSIS

The images included are frontal chest radiographs compiled by researchers from Qatar University and the University of Dhaka, in collaboration with others, and published on 25 September 2021 [20] and predominantly from the COVID-19 Radiography Database [6] and other public databases. For this study four more classes were added to the original classes to represent Tuberculosis and only the front side of images were retained. The splitter used by the authors has been kept (6,540 images in the test split). The source does not contain information about the training or validation size.

TABLE 1: Class Distribution and Training Class Weights

Class	**Label**	**Images**	**Weight**
COVID-19	0	4,189	1.5613
Lung Opacity	1	6,012	1.0873
Normal	2	10,192	0.6414
Viral Pneumonia	3	7,397	0.8837
Tuberculosis	4	4,897	1.3349
Total	—	32,687	—

The ratio for the largest size class to the smallest size class is approximately 2.4:1. The listed weights correspond to the inverse-frequency weighting, N/(K·nk), where N = 32,687, and K = 5, computed from the counts in the full data set (different by 0.0007 from the inverse-frequency weights computed from the counts in the training set; this should be checked against the source data). These figures are shown after shuffling and batching the classes and representative images from the COVID-19 after shuffling and batching at 224×224×3 are shown in Fig. 2.



Fig. 1. Mini-batch distribution

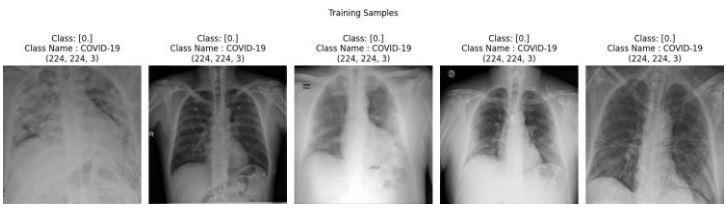


Fig. 2. These are 224×224×3 images from the training set that have been loaded by the data pipeline

The mean pixel-intensity histograms and boxplots for each class (Fig. 3) were examined. The distribution of COVID-19 was broadly single peaked, with Lung Opacity and Normal being almost indistinguishable and therefore the most difficult to distinguish; The distribution of Viral Pneumonia was mixed, with both typical and atypical patterns; The distribution of Tuberculosis was two intensity peaks. Moderate imbalance, Lung Opacity/Normal overlap, heterogeneous sources and acquisition equipment, using text markers and positioning tags as a shortcut {C("degrave")} and moderate data size for data-hungry transformers are challenges.

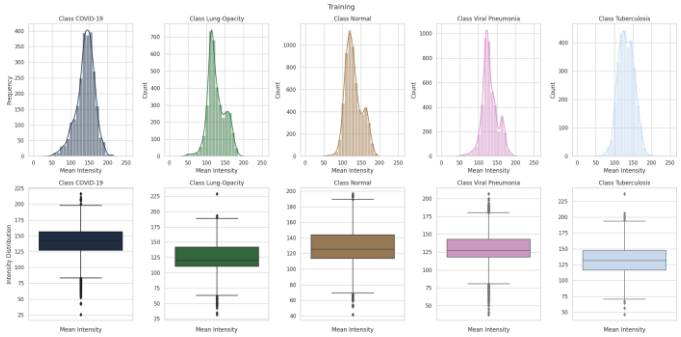


Fig. 3. Training set per class mean pixel-intensity histograms and boxplots.

IV. PROPOSED METHODOLOGY

The workflow (Fig. 5, schematic of the documented pipeline) is executed from the dataset to the preprocessing, a tf.data input pipeline, model-specific training, test-set evaluation and interpretability analysis.

A. Preprocessing

There are five stages to the pipeline, each designed in a separate notebook. (2) Zoom: a factor of 0.83 crops peripheral artefacts and scanner borders, standardising the region of interest, but a risk is clipping peripheral anatomy, the source mentions gradient activations at borders that might correlate to zooming. The choice of CLAHE clip limit and grid size was also done visually as larger grid sizes made the local noise more prominent (Fig. 4) and clip limit 2 with grid size 15 was found. (3) Sharpening: five kernels were compared and the basic kernel with centre weight of 5 and four connected neighbouring weight of -1 was selected, sharpening can also highlight noise. (4) Rescaling and (5) zero-centring:

$$I_{norm}(x,y,c) = I(x,y,c)/255 \quad (1)$$

$$I_{zc}(x,y,c) = I_{norm}(x,y,c) - \mu \quad (2)$$

where I is the intensity of channel c at pixel (x, y) and μ is a mean value. The intensities were normalized to a range, which is not legible in the provided text, and then subtracted by the mean (assigned the value μ, omitted in the provided text, which may be per-channel, global, or training-set mean); this divisor (255) and the range [0, 1] and the origin of μ (per-channel, global, or training-set mean) are therefore assumptions to be verified against the code. Average-intensity distribution was used to check the pipeline stages. A tf.data pipeline was set up to shuffle, batch, one-hot encode the labels, map them, and then preload the final tensors.

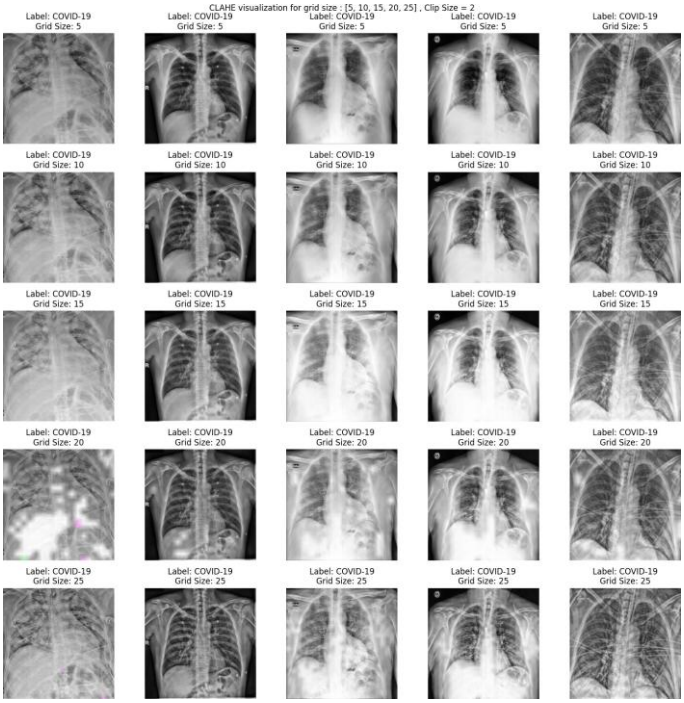


Fig. 4. CLAHE grid-size sweep (5–25, clip limit 2) on COVID-19 samples; grid size 15 was selected

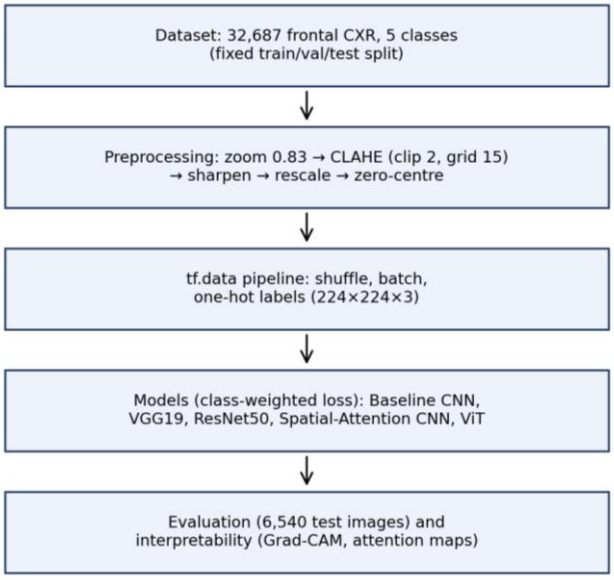


Fig. 5. Schematic of the documented workflow

B. Problem formulation and loss

Let x be a preprocessed CXR and C = {c0,…,c4} the labels COVID-19, Lung Opacity, Normal, Viral Pneumonia and Tuberculosis. The five probabilities are produced by each model:

$$p = f_\theta(x), \quad \Sigma_{k=1}^{5} p_k = 1 \quad (3)$$

$$L_{CE} = -\sum_{k=1}^{5} y_k \log(p_k) \quad (4)$$

$$L_{WCE} = -w_y \log(p_y) \quad (5)$$

In this case, yk is the one-hot ground truth, y the true class, py the predicted probability of the true class and wy the class weight of the true class from Table II. The source does not specify whether global class weights were used for loss in all models or weights were used per sample. The avoidance of oversampling was due to the possibility of leakage of duplicate images between splits.

C. Model architectures

Baseline CNN:The model is a stacked Conv2D/activation/max-pooling/dropout network with a softmax output and 104,038,981 parameters (this is actually the source text that wrote "billion", which is the number of parameters, thus the error), which was primarily used to validate the data pipeline. There is no documentation in the provided source of how to set up the layer.

VGG19: Two dense layers plus a five-class softmax (38,927,429 parameters) attached to a 19-layer VGG network [21] that is pre-trained on ImageNet. Stage 1 - train the new head with the base frozen; stage 2 - unfreeze the base at the lower learning rate (partially or fully, depending on the source; the VGG19 curves refer to training the full network with a low learning rate).

ResNet50: The 50-layer residual network [22] with ImageNet weights, global average pooling and a single dense softmax layer (23,597,957 parameters; 23,544,837 trainable), with the same two-stage protocol. A residual block is a computational operation that calculates where the identity addition needs to be compatible, and shortcuts to projections when the dimensions differ.

TABLE 2: Model Comparision

Model	**Weights**	**Attention**	**Params**	**Strategy**	**Explainability**
Baseline CNN	Random	None	104.0 M	Scratch	None reported
VGG19	ImageNet	None	38.9 M	2-stage	Grad-CAM, Guided Grad-CAM
ResNet50	ImageNet	None	23.6 M	2-stage	Grad-CAM, Guided Grad-CAM

D. Interpretability

Grad-CAM [18] uses a map and global-average-pooling gradients to calculate importance weights αkc for a class score yc, for a feature map. Guided Grad-CAM performs the map element-wise multiplication by guided backpropagation. It was attached to the VGG19 and ResNet50 models, and weights for the attention were obtained from the custom CNN gates and the ViT encoder blocks. These maps provide information on the sensitivity of a model, but they do not provide evidence of the clinical validity or causality.

V. IMPLEMENTATION AND EXPERIMENTAL SETUP

Code was written in Python 3 with TensorFlow/Keras, tf.data, OpenCV, NumPy, Pandas, Matplotlib and Seaborn in Jupyter notebooks. A random seed of 123 was set for NumPy and TensorFlow. All models will use the same fixed split of 6,540 images—this is the evaluation set. The configuration is shown in Table IV; where not available, it is marked not estimated.

TABLE 3. Experimental Configuration

Item	**Value**
Operating system / CPU / GPU / RAM	Not documented
Python / TensorFlow / OpenCV / NumPy / Pandas versions	Not documented (Python 3, TensorFlow/Keras)
Environment	Jupyter Notebook
Input shape	224×224×3 (Fig. 2 caption)
Classes / seed	5 / 123
Split	Fixed dataset-authors' split; test = 6,540
Class weights	Table 1
Batch size, optimizer, learning rates, epochs	Not documented
Fine-tuning layers, early stopping, checkpoint rule	Not documented (checkpoint criterion discussed in Sec. VI)
Dropout rate, data augmentation	Not documented
Training time / memory	Not documented

Training protocol. VGG19 and ResNet50 have two phases (frozen base phase training, then unfreezing and training at a lower learning rate). The custom CNN and ViT were trained from scratch. Validation data were used to select the model, and the test split was used for final

reporting; no indication of whether the test data influenced selection of the model.

VI. RESULT & COMPARATIVE ABALYSIS

Weighted averages weight by support and macro averages weight equally. ROC-AUC is threshold independent, top 2 accuracy is the percentage of images with the correct class in the top 2 probabilities. The source reports per-class values of macro and weighted averages but it is not reported which kind of averaging was performed for summary precision, recall, F1 and AUC (Table 4) and the scheme for multiple-class (one-vs-rest or one-vs-one) averaging of ROC-AUC is not documented.

TABLE 4. Test Set results as reported in source

Model	**Loss**	**Acc.**	**Prec.**	**Rec.**	**F1**	**AUC**	**Top-2**
VGG19	0.1598	0.9453	0.9463	0.9434	0.9449	0.9949	n/a
Custom CNN	0.2087	0.9700*	0.9285	0.9209	0.9247	0.9930	n/a
ResNet50	0.2229	0.9367	0.9398	0.9350	0.9374	0.9907	n/a
ViT	0.6624	0.7865	0.8116	0.7615	0.7857	0.9513	0.9365
Baseline CNN	n/a	0.92	0.92	0.92	0.92	n/a	n/a

Discrepancy check (unresolved). The source claims a test accuracy of 0.9700 on a custom-CNN with a 6,540 image test set, but according to its classification report, the test set is 0.92 on the same set of 6,540 images and recall (0.9209) and F1 (0.9247) are also 0.92, not 0.97. In the case of single label multiclass data, the weighted-average recall is equal to the accuracy, meaning that 0.97 contradicts 0.9209. This cannot be verified by the supplied files, as the source speculates that different checkpoints are to blame, and it would require the prediction outputs and training logs. This implies that no inference is drawn that relies on 0.97 and cross-model comparisons are made using macro F1 and ROC-AUC (Fig. 6). Other differences in the source include that the abstract attributes accuracy 0.945 to the custom CNN and AUC 0.996/0.997 to the custom CNN and ResNet50, do not match the values reported in Table 4; and that the reported parameter reduction is “about 40 times” less than that reported in the count (38,927,429/2,543,845 = 15.3).

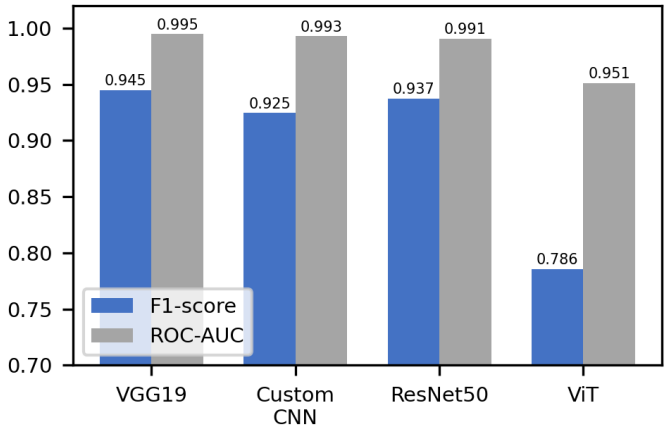


Fig. 6. Macro F1 and ROC-AUC of the four models in Table V

TABLE 5. Per Class F1 score from the classification

Class	**Base**	**VGG19**	**ResNet50**	**Custom**	**ViT**
COVID-19	0.91	0.95	0.94	0.90	0.63
Lung Opacity	0.84	0.90	0.90	0.86	0.73
Normal	0.90	0.94	0.94	0.92	0.81
Viral Pneumonia	0.96	0.97	0.95	0.97	0.88
Tuberculosis	0.97	0.98	0.94	0.97	0.81

Transfer learning achieved the highest F1 values (VGG19, ResNet50), with the custom CNN with 2.5 M-params obtaining a macro F1 of 0.925, which was within ~2 points of VGG19 and ~1 point of ResNet50 and the lowest per-class F1 for Lung Opacity (0.86) and COVID-19 (0.90). The from-scratch ViT was obviously the weakest, with the precision at COVID-19 being 0.59, and the recall being 0.67; the top-2 accuracy was 0.9365, and ROC-AUC was 0.951. This is aligned with reports that ViTs require large amounts of data, pretraining and regularisation {C("steiner")}, {C("shamshad")} and is consistent with the present study in that it was not causally tested. There was no pretrained ViT run here, and thus no conclusion about CNN vs transformer beyond this setting is warranted.

The VGG19 confusion matrix is presented here as an example in Fig. 7. The source can be used to see that Lung Opacity/Normal confusion is the largest error source for the CNN-family models, as is the case for the near-identical intensity distributions, and the lack of TB/COVID-19 confusion for all models, while ViT confusion of COVID-19 with several classes; the other matrices are not reproduced here. VGG19 curves (Fig. 8) show that the training accuracy increases and the

validation accuracy is not shown with a significant increase, which suggests mild overfitting during the fine tuning, with a similar trend reported for ResNet50. The source mentions the noisy initial validation curves for the custom CNN which merge together, which is not shown in the source.



Fig. 7. Test-set confusion matrix of VGG19

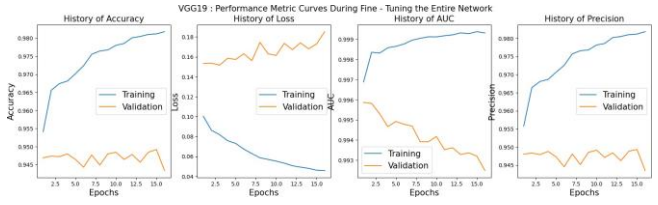


Fig. 8. VGG19 accuracy, loss, AUC and precision curves during full-network fine-tuning

A. *Explainability results*

Qualitative results from the source were: Grad-CAM maps of ResNet50 were able to identify regions of the chest where the disease was present, but also were able to identify image borders and noise; VGG19 activations were more affected by acquisition artefacts; the custom CNN was able to attend to the lungs in deeper blocks, but the attention focus was not focused on the chest area in general; ViT attention focused inconsistently on the patches, sometimes including irrelevant patches. Grad-CAM, Guided Grad-CAM and attention-map images were not included in the provided file and are therefore not reproduced and these descriptions have not been verified independently. Plausible heatmaps do not imply the use of clinically relevant features, and saliency methods can localise poorly compared to radiologists {C("saporta")}.

B. *Benchmark comparison*

The project is compared with similar studies in Table 6. The numbers are not directly comparable because: datasets, label sets, splits and metrics vary, and none of the published models were re-run here.

Table 6. Contextual comparison with published studies

Ref.	Dataset / task	Model	Published result	Relation
[5]	COVQU-20, 3-class	Six CNNs + enhancement	Acc. 96.29% (gamma)	Same dataset lineage; fewer classes
[13]	CXR, COVID-19/pneumonia/normal	xViTCOS (pretrained ViT)	Not extracted	Closest ViT study; uses transfer
[14]	CXR, COVID-19 detection	ViT + radiomics	COVID-19 sensitivity 0.99	Binary framing
[7]	COVID-19 CXR	Attention VGG-16	Not extracted	Attention-CNN analogue

VII. LIMITATIONS AND FUTURE WORK

Limitations: Only evaluated based on one dataset; Sources are heterogeneous and there could be artefacts; Moderate imbalance; Potential mild overfitting; One run, no variance estimate; Hardware, hyperparameters, and several architecture details were not documented; The accuracy discrepancy is unresolved; No clinician evaluation. Large and multi-site datasets, pretrained or hybrid CNN–ViT models, ensembling, calibration and statistical testing, further backbones, complementary explanation tools (e.g., SHAP or Integrated Gradients), validation and compression by radiologists for point-of-care use are potential future work.

VIII. CONCLUSION

Pretrained VGG19 and ResNet50 and a compact spatial-attention CNN achieved macro F1 scores of 0.925–0.945 and ROC-AUC scores of 0.991–0.995, whereas a from-scratch ViT achieved 0.786 and 0.951 on a fixed split. In interpretation analysis, there was some indication of the use of non-anatomical artefacts. This is an image classification study, not a diagnostic system; external validation and prospective evaluation would be needed before adopting for clinical use and the discrepancies reported here should be resolved.

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